

Practical No.10

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Objective:

1. Create a DataFrame for 5 students with columns: Name, Age, Grade, City.
2. Filter students with Age > 20 and Grade > 85.
3. Add a new column Passed which is True if Grade >= 50.
4. Sort the DataFrame by Grade in descending order.
5. Group students by City and calculate the average Grade.

Python Code:

```
In [5]: import pandas as pd
import numpy as np
```

1) Creation of Dataframe

```
In [6]: students_data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva'],
    'Age': [22, 19, 23, 21, 20],
    'Grade': [88, 76, 90, 65, 95],
    'City': ['New York', 'Chicago', 'New York', 'Chicago', 'Boston']
}

students_df = pd.DataFrame(students_data)
students_df
```

```
Out[6]:
```

	Name	Age	Grade	City
0	Alice	22	88	New York
1	Bob	19	76	Chicago
2	Charlie	23	90	New York
3	David	21	65	Chicago
4	Eva	20	95	Boston

2) Filter students with Age > 20 and Grade > 85.

```
In [7]: filtered_df = students_df[(students_df['Age'] > 20) & (students_df['Grade'] > 85)]
filtered_df
```

```
Out[7]:
```

	Name	Age	Grade	City
0	Alice	22	88	New York
2	Charlie	23	90	New York

3) Add a new column labelled "Passed" which is True if Grade >= 50.

```
In [8]: students_df['Passed'] = students_df['Grade'] >= 50
students_df
```

```
Out[8]:
```

	Name	Age	Grade	City	Passed
0	Alice	22	88	New York	True
1	Bob	19	76	Chicago	True
2	Charlie	23	90	New York	True
3	David	21	65	Chicago	True
4	Eva	20	95	Boston	True

4) Sort the DataFrame by Grade in descending order.

```
In [9]: sorted_df = students_df.sort_values(by='Grade', ascending=False)
sorted_df
```

```
Out[9]:
```

	Name	Age	Grade	City	Passed
4	Eva	20	95	Boston	True
2	Charlie	23	90	New York	True
0	Alice	22	88	New York	True
1	Bob	19	76	Chicago	True
3	David	21	65	Chicago	True

5) Group students by City and calculate the average Grade

```
In [10]: grouped_df = students_df.groupby('City')['Grade'].mean()
grouped_df
```

```
Out[10]: City
Boston      95.0
Chicago     70.5
New York    89.0
Name: Grade, dtype: float64
```

Interpretation:

1. As we can see Data Frame was created with name 'students_df'.
2. Filtered students with Age > 20 and Grade > 85 whose are 'Alice' and 'Charlie'.
3. Added a new column labelled "Passed" which are 'Alice', 'Bob', 'Charlie', 'David' and 'Eva'.
4. Sorted the DataFrame by Grade in descending order as 'Eva', 'Charlie', 'Alice', 'Bob' and 'David'.
5. The average Grades of students grouped by City Boston ,Chicago and New York are 95.0 ,70.5 and 89.0 respectively.